

VIJAY PRASAD

Chief Technology Officer · Robotics & Drone Systems Architect · AI Developer · Security-by-Design

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EXECUTIVE SUMMARY

CTO at DLD Hariyalikart Pvt. Ltd. and B.Tech (CSE — AI & ML) student with hands-on experience building AI-powered agricultural robotics, secure drone ecosystems, and full-stack web platforms. Proven leader in India's IIT entrepreneurship and technical circuits (E-Cell IIT BHU, Techkriti IIT Kanpur, Technex IIT BHU). Passionate about bridging hardware agility with software integrity to solve real-world challenges at scale.

EDUCATION

B.Tech in Computer Science & Engineering (AI & ML)

2025 – 2029 · Expected Graduation: 2029

- Specialisation: Artificial Intelligence & Machine Learning
- Relevant coursework: Data Structures, Machine Learning, Deep Learning, Embedded Systems, Computer Networks, Cybersecurity

PROFESSIONAL EXPERIENCE

Chief Technology Officer

DLD Hariyalikart Pvt. Ltd. · 2024 – Present

- Spearhead end-to-end technical vision for an agri-tech startup, defining architecture for AI-driven supply chain and precision farming platforms.
- Designed and deployed AI-powered agricultural robotics for autonomous crop monitoring, integrating Python, C++, Arduino, and Raspberry Pi.
- Built a secure drone ecosystem with hardened firmware, encrypted OTA update architecture, and edge-level threat modeling.
- Led cross-functional engineering team; managed product roadmap from prototype to production-ready deployment.
- Developed Krishi Mitra AI — a Gemini AI-powered crop management companion (Next.js) — and Smart Monitoring Systems (TypeScript/React/NodeJS) for real-time hardware telemetry.

Campus Ambassador

E-Cell IIT BHU · 2025 | Techkriti IIT Kanpur · 2025 | Technex IIT BHU · 2026 | Techfest IIT Bombay · 2026

- Represented four of India's premier IIT-level technical and entrepreneurial festivals — IIT Bombay, IIT BHU, and IIT Kanpur — across consecutive years.
- Drove regional student engagement, coordinated robotics/AI workshops, and mentored aspiring founders in the startup ecosystem.

KEY PROJECTS

AI-Powered Agricultural Robotics

Python · C++ · Arduino · Raspberry Pi · MySQL

Autonomous system for precision farming and crop monitoring; integrates computer vision for real-time field analytics.

Secure Drone Ecosystem

C · Security Architecture · Firmware · Encryption

Hardened firmware and encrypted OTA update pipeline for industrial drones; secure-by-design from boot chain to communication layer.

Krishi Mitra AI

Gemini AI · Next.js · Agriculture Tech

AI-powered agricultural companion providing real-time crop insights. Live: krishi-mitra-ai.vercel.app

Smart Monitoring Systems

TypeScript · NodeJS · React · Nginx · SQLite

AI-driven analytics platform for real-time hardware telemetry and system health dashboards.

Campus Lens

Next.js · Tailwind CSS · React

Web platform for campus navigation, student hubs, and real-time community engagement. Live: campus-lens-v2.vercel.app

Ignizia Tech

Web Development · UI/UX · Cloud Solutions

Tech development and solutions platform offering custom web engineering. Live: ignizia.in

TECHNICAL SKILLS

Languages	Python, C, C++, JavaScript, TypeScript, HTML5, MySQL
Frameworks & Platforms	Next.js, React, NodeJS, Arduino IDE, Raspberry Pi, Nginx, Apache
AI & Cloud	Gemini AI, Vertex AI, Google Cloud, Salesforce, Vercel
Hardware & Embedded	Firmware Development, Secure Boot, OTA Architecture, Circuit Design, PCB Prototyping, CAD/SolidWorks
Security	Threat Modeling, Encryption at the Edge, Secure Communication Protocols, InfoSec (ISEA Certified)
Tools	Git, Docker, Linux, VS Code, Autodesk Fusion 360, AutoCAD, Geomagic Design X

CERTIFICATIONS

- Autodesk Certified Professional — Fusion 360 & AutoCAD
- SolidWorks Certified Professional — Mechanical Design & Manufacturing
- Geomagic Design X Expert — Reverse Engineering Workflows
- InfoSec Awareness (ISEA / MeitY) — Cert ID: MeitY/ISEA/WCHP/029371
- GUVI HCL — C Programming & OOP using C++
- 11x Udemy Certifications — AI, Robotics, Web Development & Security domains
- International Physics Quiz (IPhYC) — 2025

HONORS & RECOGNITION

- Technical Innovator Award — CTO @ HariyaliKart for agricultural transformation through scalable AI.
- Robotics Systems Architect — Integration of AI-powered solutions in agricultural robotics.
- Elite Ambassador — Represented E-Cell IIT BHU (2025), Technex IIT BHU (2026), Techkriti IIT Kanpur (2025), and Techfest IIT Bombay (2026).